

6 Fig. 6.1 shows part of a bridge rectifier circuit that can be used for rectification of an alternating input voltage V_{IN} .

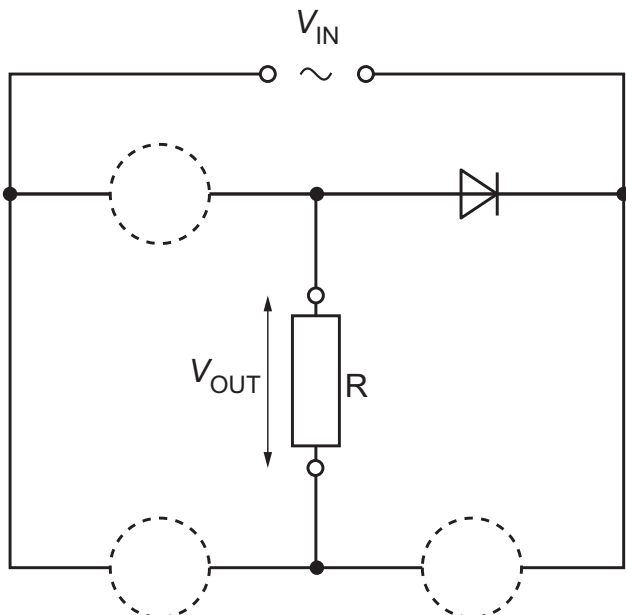


Fig. 6.1

The circuit contains four diodes, one of which is shown.
The rectified output voltage V_{OUT} is applied across load resistor R .

- (a) (i) State what is meant by rectification.
- [1]
- (ii) State the name of the type of rectification produced by a bridge rectifier circuit.
- [1]
- (iii) Complete the circuit in Fig. 6.1 by drawing the three missing diodes inside the dashed circles. [2]

(b) The input voltage varies with time t according to the equation

$$V_{IN} = 34 \sin 18t$$

where V_{IN} is in V and t is in s.

- (i) Show that the period of the input voltage is 0.35 s.
- [2]
- (ii) Calculate the root-mean-square (r.m.s.) input voltage.

r.m.s. voltage = V [1]

(iii) On Fig. 6.2, sketch the variation of V_{OUT} with t from $t = 0$ to $t = 0.35$ s.

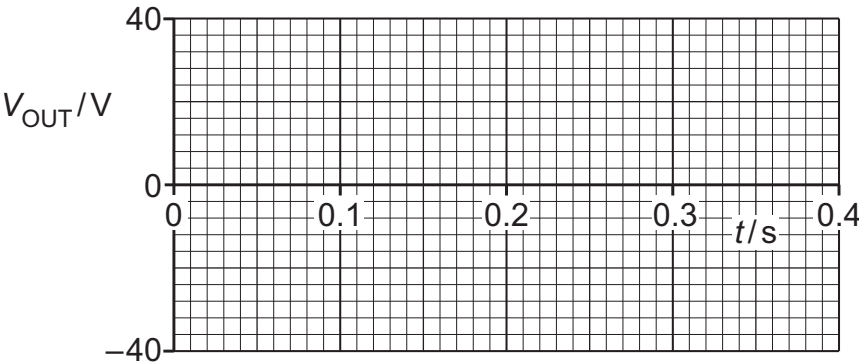


Fig. 6.2

[3]

(c) Resistor R has a resistance of $56 \text{ k}\Omega$. A capacitor of capacitance $12 \mu\text{F}$ is connected into the circuit of Fig. 6.1 in order to smooth the output voltage.

- (i) On Fig. 6.1, draw the capacitor correctly connected into the circuit. [1]
- (ii) Calculate the time constant of the smoothing circuit.

time constant = s [2]

(iii) During each discharge cycle, the time for which the capacitor is discharging is 0.14 s.

Determine the minimum value of the smoothed output voltage.

minimum voltage = V [2]